

Assisting Learning and Insight in Design Using Embodied Conversational Agents

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Abstract Reflective practices correlate with both insight in engineering design projects (Currano 2015) and learning in engineering education (Chew et al. 2016). Schön (1992) observed that designers engage in a reflective conversation with design materials as they sketch and prototype their ideas. However, students can struggle when learning domain knowledge and design practices simultaneously. We hypothesized that (1) making the conversation between student designers and their prototyping materials literal and explicit, rather than metaphorical, can assist in reflection, and (2) the effect can be enhanced by embodying a conversational agent within the prototype itself. We developed an embodied conversational agent as a tool to elicit reflection, and lead to greater insight and learning, during a hands-on mechatronics prototyping and design activity. In addition to guiding learners through a tutorial, the agent answers questions, offers comments, and asks deep reasoning and generative design questions, which correlate with convergent and divergent phases of the design process (Eris 2003). We compare learners' gained knowledge, performance, and reactions in conditions with or without a conversational agent, and using an embodied or non-embodied agent, to evaluate the impact of (1) conversational communication, and (2) physical embodiment, on learning and insight. In this chapter we describe the background and rationale for our study, details of the study design, and preliminary insights from our pilot data.

1 Context & Human Need

In *The Reflective Practitioner*, Donald Schön developed the concept of *reflection-in-action*—a process of in-the-moment reevaluation and redirection of design activity—in part through observation and interviews with student and practicing designers engaged in open-ended architectural design tasks (Schön 1983). Over the last decade of

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working with engineering design students, we have observed that student learning and attention are often split between two contexts: (1) the domain knowledge that underlies the design task, and (2) the process of design itself. Being students, these novice designers are neither formally trained in, nor naturally skilled at, reflecting on their process in the midst of action. They learn how to reflect productively through practice and a concerted effort.

Seeking help and consulting with design educators, practitioners, or domain experts can support students' domain and process knowledge, but not necessarily their reflective practices. As a result, design students often lack direction, cognizance of, and experience with reflective practice, and thus may struggle in drawing broader insights during their design projects. In addition, the search for these different types of knowledge entails interrupting the flow of design activity to identify and contact the requisite expert(s), elicit information, and then synthesize learnings and how to apply them. This switching between contexts can be a further challenge to reflection, but it offers an opportunity to develop tools to elicit and guide reflection on process and content that can be applied during design activity.

2 Prior Work & Background

In extending his work on reflection-in-action, Schön (1992) observed that designers learn and gather information, identify patterns, construct meaning, and assess qualities in a process of *seeing-moving-seeing* through reflective engagement with various media, including sketches, prototypes, and components. He called this a “*conversation with the materials of a design situation*,” highlighting the emergence of unintended effects of moves (e.g., changes, additions, reconfigurations), which result in the evolution of both the design itself, and the designer's intentions.

2.1 *Design is a Socio-Technical Process*

While Schon's and other works have recognized the integral role visual thinking plays in design (McKim 1972), more recent work also shows the importance of verbal, spoken conversation in helping designers distill insights from and about their design tasks (Currano 2012). Sonalkar et al. (2016) developed a tool (the Interaction Dynamics Notation tool, or IDN) to help identify patterns of team interaction—which were primarily conversational—and showed that these interactions correlate with creativity measures in concept generation design outcomes. This makes sense, given the socio-

technical nature of the design process (Cross & Cross 1995), and particularly for teams actively communicating with each other and interacting with shared prototypes.

Cross' seminal research on the intersection of the social process of design with the technical and cognitive processes was based heavily on observation and analysis of the team conversations during their design activity.

2.2 *Question-Asking in Design*

Also focusing on design processes, Dym et al. (2005) characterized design, in part, by the focus on asking and answering questions. In particular, Eris (2003) identified specific types of questioning behavior that designers engage in—deep reasoning questions (DRQs) and generative design questions (GDQs)—and found that these correlate respectively to convergent and divergent phases of design.

Choi et al. (2005) observed that student designers found it difficult to generate quality questions. However, using a text-based scaffolding tool to support peer-questions, they found that simple adaptive questioning prompted reflection, facilitating metacognition, and enhancing learning during online discussions.

Schön's *seeing-moving-seeing* process suggests that design learning happens through a reflective hands-on engagement with design materials, and that the materials used by designers in their work serve two purposes: (1) as components of the prototyping and sketching process, and (2) as media for a reflective learning process. Additionally, the research in question asking and the social process of design together indicate benefit in not only asking questions, but doing so within the context of spoken conversation.

We therefore believe that more integrally combining the two roles of design materials by *literally* endowing the materials with the capabilities of conversing about the design process and progress, and adaptively asking convergent and divergent questions, will enhance design and learning processes, especially for novice designers.

2.3 *Physically Embodying Materials*

The concept of embodying materials and endowing them with the ability to verbally engage designers has been explored in recent studies. In the context of an electronics prototyping activity, Jung et al. (2014) tested whether an agent that was embedded in or external to the physical material affected students' perceptions and learning outcomes, while also varying the agent's expressions of social interest toward the participant and activity. They found that an embedded agent lowered perceptions of task

stress, and an interested agent was more likable and socially present. The influence of either condition on learning outcomes was mixed, but suggested relevance to students' prototyping confidence and performance.

Using a similar prototyping activity, Martelaro et al. (2016) tested how varying the agent's expressivity and vulnerability influenced students' perceptions of trust, disclosure, and companionship. They found that increased vulnerability led to higher ratings of trust and companionship, and increased expressiveness produced higher ratings of disclosure, with trust mediating vulnerability and companionship.

These studies directly inform our guiding questions, research context, and implementation. While their hypotheses focus on affective expressions of emotion and attention, and their measures emphasize social relations, our approach addresses a gap in understanding around the role of (a) interactive questioning conversation in an agent-based learning context, and (b) a teaching agent's physical embodiment, in supporting learners' abilities to reflect and build insight during a design task.

3 Research Approach

3.1 Research Questions

Bringing these 3 themes together—the role of reflection, spoken conversation, and divergent and convergent questioning—inspired us to study design learning in the more interactive and physical contexts of teaching mechatronics. Our guiding research questions are therefore:

1. Can conversational agents help novice designers reflect, develop insight, and improve learning by making their conversation with design materials more literal and explicit?
2. Furthermore, given the physicality of working with materials and prototyping, can embodying these conversational agents support and enhance reflection, insights, and learning for engineering design tasks?

3.2 Hypotheses

The aim of this work is to thoughtfully integrate conversation and question asking into the prototyping process for novice designers to enhance their learning and design outcomes. Given the social nature of design and the implicit conversations embodied

in seeing-moving-seeing that lead to the emergence of new insights in the reflective process, we hypothesize that:

1. Making the conversational layer explicit (via an agent that asks the learner questions, and answers learners' questions) will spark deeper/broader seeing and encourage more diverse moving.
2. The physicality afforded by embodying the conversational agent within the prototype being designed (e.g., as a robot) will assist novice designers' reflective process and reduce shifts in attention.

3.3 Research Setting

We have developed a conversational agent, embodied as a robot, that responds to the moves that learners make, and asks questions that elicit and guide reflection within their process, thereby helping them to learn from and discover new insights and evolve their designs. Mechatronics prototyping is an opportune context to study the use of conversational agents in the design process. It, provides a natural setting to integrate an embodied conversational agent into the prototyping materials and to test its effects on insight and learning. For example, Martelaro et al. (2016) examined human-robot conversation within a mechatronics tutorial, although they focused on the social aspects of trust, disclosure, and companionship rather than learning and insight.

An introductory mechatronics tutorial that we developed and have taught over the last several years at the ACM Conference on Human Factors in Computing Systems (Sirkin & Ju, 2014) serves as our starting point (see Fig. 1).

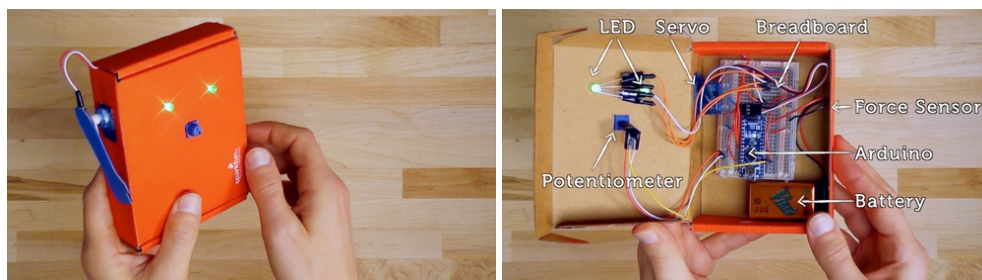


Fig. 1 Paper robot project from our conference course in electronics prototyping using Arduino.

3.4 Study Design

In an experiment employing this conversational agent (implemented using a human-in-the-loop protocol), participants follow an abbreviated version of the conference tutorial to build a small paper robot and learn basic mechatronics. The study is split into 4 sections: a pre-activity survey to evaluate participants' knowledge of and comfort with electronics prototyping, the learning tutorial, a sketched design task, and a post-activity survey to evaluate student learnings and perceptions of the agent. The entire study takes about 1 hour.

The study has a 2 (conversational vs non-conversational) x 2 (embodied vs non-embodied) design, each corresponding to one of our two hypotheses. In the conversational condition, the agent speaks in an engaging and interactive style and asks questions, while in the non-conversational condition, it communicates the same background and instructions, but speaks in a non-interactive style without asking questions. In the embodied condition, the agent refers to itself, and its voice projects from the paper robot, while in the non-embodied condition, it refers to the robot as a separate entity and its voice comes from the laptop that participants use to program the robot.

3.4.1 Conversational vs Non-Conversational

During the tutorial, both conversational and non-conversational agents offer guidance on how to assemble the robot. They provide information such as component names and uses (e.g., *"A potentiometer is a variable resistor. When you turn the knob, the resistance changes between the middle pin and the outer pins."*), and they also offer suggestions about how to correctly connect components together (e.g., *"You will need to add a resistor first, to limit the current in my/the LED to a safe value. Otherwise, it might burn out."*)

The conversational agent uses a slightly less formal speaking style appropriate to spoken conversation, while the non-conventional agent uses a style appropriate to written instructions. In addition, the conversational agent interjects DRQs (e.g., *"A photocell is an ambient light sensor. What do you think makes its resistance change?"*) and GDQs (e.g., *"Can you tell me a few different things you could use a servo to do?"*) to inspire convergent and divergent oriented reflection.

In both conditions, participants can ask the agent questions and receive answers about components or instructions, or ask for statements to be repeated, to ensure that they have adequate access to the information presented. However, in the conversational condition the agent tells participants *"You can ask me questions at any time."* whereas in the non-conversational condition, it instructs them to direct their questions to the study administrator.

3.4.2 Embodied vs Non-Embodied

During the tutorial, both embodied and non-embodied agents provide the same information but in a different manner. The embodied agent speaks in a self-referential style (e.g., *“Hello, my name is Robbie. I’m a robot and today you are going to build me.”* and *“You can plug my potentiometer into the blue sockets of my breadboard.”*), in contrast to the non-embodied agent, which refers to the robot as a separate entity (e.g., *“Today you will be building a small paper robot.”* and *“Look for the 10K Ohm resistor on the robot’s breadboard.”*). In both cases, the agent speaks with the same synthetic male voice, raised in pitch and slowed in rate 15% from its default setting, to better suggest it as being a rough prototype—more youthful and less authoritative.

At the same time, the embodied agent’s voice emanates from a small Bluetooth speaker mounted to the same physical platform as the robot’s other components (see Fig. 2), while the non-embodied agent’s voice originates from the speakers of the laptop computer that participants use to edit Arduino sketches and upload them to the robot.

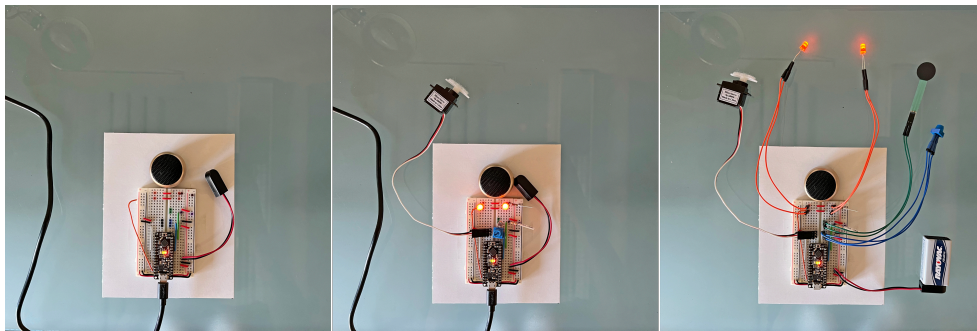


Fig. 2 Paper robot project from the current study, showing the mechatronics platform in 3 states: (left) as participants first encounter it, (center) assembled to complete its functionality, and (right) with components moved off-board and ready to be inserted into the cardboard box in Fig. 1.

3.4.3 Designing a New Robot

After participants complete the tutorial, they complete a task to design one or more different robots using the same types of components, thinking about what they would want their robot to do, and how it might do those things. Participants are encouraged to think about components at different scales, to incorporate as many components as they like, and to create additional robot designs if they want. They are also asked to make sketches and notes as they work out their ideas. To encourage novelty, our instructions state that we will evaluate all participants’ designs and rank them based on creativity, but we will not evaluate their drawing skills, just how they reimagine or reapply the tools they learned in the tutorial. The design task is timed and takes 15 minutes.

3.5 Data, Measures & Analysis

Data are collected from 3 sources: (1) physical actions and conversation by participants, (2) sketches from the robot design task, and (3) pre- and post-activity questionnaire and post activity interview responses. Measures include the depth and breadth of design ideation, instances of reflective practice (manifest as physical or verbal *conversations with materials*), locus of attention, and learning, across study conditions.

3.5.1 Physical Actions & Conversation

During the tutorial and robot redesign task, we video-record participants' physical actions and spoken interactions with the agent for coding and analysis (Heyman et al. 2014) to identify behavioral patterns and understand levels of engagement across conversational and embodiment conditions. For coding, 2–3 researchers codevelop an evaluation framework based on a subset of the data, using Cohen's kappa (Lombard et al. 2002) to check inter-rater reliability, and then they apply it to the remaining participant interactions. An initial analysis of pilot participant data will focus on two alternatives as proxies for reflection and learning: 1) actions such as asking follow-up questions and manipulating materials while ideating or answering questions, and 2) performance measures such as task completion time or number of mistakes.

3.5.2 Robot Designs

We also analyze the notes, descriptions, and sketches resulting from the robot design task, focusing on (1) flexibility, through changes made to the robot's form, functions, context of use, or the means through which these new functions are accomplished, and (2) fluency, through participants' ability to apply this knowledge to create a breath of distinct designs.

3.5.3 Questionnaire

We query participants through pre- and post-task surveys and a brief interview, first about their prior domain knowledge and design process experience, and then about their design intentions, learnings, insights, and reflections. Questions include novel and standardized measures, such as hedonic and pragmatic user experience scales (UEQ-S) (Schrepp et al. 2017). Questionnaire data will be fit using generalized linear or mixed effects models, which offer flexibility in data form and conditions, and support random effects for participants and conditions, potentially lowering the chance of type 1 error (Boisgontier & Cheval 2016). We also include questions to confirm whether participants recognize and respond to the manipulations within their study

condition, including direct questions and measures of social presence such as copresence, involvement, and engagement (Biocca et al. 2003).

4 Preliminary Findings

4.1 *Revisions Made While Piloting*

4.1.1 Clarifying Conditions

In pilot testing, we found that having the agent ask intermittent check-in questions between tasks in the instructions such as “*Are you following along so far?*” encouraged participants to remain engaged in the conversation and speak up at other times. We also added small interjections like “*Ok, great!*” and “*Good work!*” to acknowledge participants’ statements and answers and comply with basic principles of politeness in conversational interaction. These additions help separate conversational and non-conversational conditions.

Also, to be sure participants correctly perceive the embodied and non-embodied conditions, we added three checks. These were 1) a manipulation check asking participants where the agent’s voice seemed to come from (robot vs computer vs somewhere else), 2) verification of how extensively they talked with the agent (not at all, somewhat, a lot), and 3) identification of the entity with whom it felt like they were speaking (the robot they were building, the computer, or something else). |

4.1.2 Clarifying Instructions

During the tutorial task, participants were confused by the presence of lettering and numbering on both the sockets of the breadboard and the pins of the Arduino. Updating the early portion of the tutorial to note these distinctions resolved their uncertainty. Participants also missed hearing or sometimes forgot component names. Labeling the components in addition to describing them helped participants identify them more easily. Finally, participants often got distracted when finding, examining, and assembling components. By better separating and reordering operations they could more easily keep focused on one task at a time.

4.2 Observations of Reflective Behavior

In the conversational conditions, we encourage reflection by asking questions about components just after participants pick them up or plug them in. Similarly, we ask questions about programming component behavior just after they look at the Arduino code. In the non-conversational conditions, we encourage reflection by telling participants to notice or observe what happens after they manipulate components or upload the Arduino code.

Pilot data in the conversational embodied condition demonstrates that when participants are asked what alternative things a servo might be used for, they often hold the components up and test how they work, squeezing the force-sensing resistor and watching the servo rotate a few times, which then provides inspiration for their answers. Likewise, when they are asked how else they might modify the ambient light coming from a photocell, they often wave their hands over the sensor and observe its effect. These examples show that asking questions during the tutorial can help promote learners engaging in reflective conversations with the materials of a design situation, and that we can observe the process of seeing-moving-seeing within these conversations.

We also have preliminary evidence that speaking reflections out loud, rather than reflecting silently, can help learners think about components and connections, and think through their answers to questions, more effectively. We observed some instances of participants reflecting aloud about various instructions, wondering for example, how they can know whether the Arduino (or some other source) is powering the LEDs, or if it would be safe to plug in a second LED without adding a second resistor. Likewise, arm and hand gestures may help support the ideation process, as we observed participants gesticulating as they thought (and spoke) through their answers to GDQs, which sometimes inspired additional ideas.

4.3 Observations of Robot Redesign

Participants took very different approaches to creating and describing their designs. One sketched several robots, with minimal text, keeping the tutorial robot's form, but changing which components were used for the eyes, nose, and arms, and adding other features such as a mouth or ears (see Fig 3). Another used almost exclusively text, focusing on a single component in each design, stating a task that a robot would use it to do (e.g., kick a ball or erase a whiteboard) without describing other aspects of the robot (see Fig. 4). Others took a more needs-oriented approach and used a combination of sketches and accompanying text to describe how a new device might use one

or two components to solve an everyday problem (e.g., call a waiter in a restaurant or keep a laptop from going to sleep) (see Fig. 5).

The most creative of the designs were the ones that focused on needs. These explored how components might be used in a new way, and a specific setting, to solve a real-life problem participants had experienced.

Fig. 3 One pilot participant's robot redesign sketches focused on alternative configurations, keeping the overall design but changing or adding components and features.

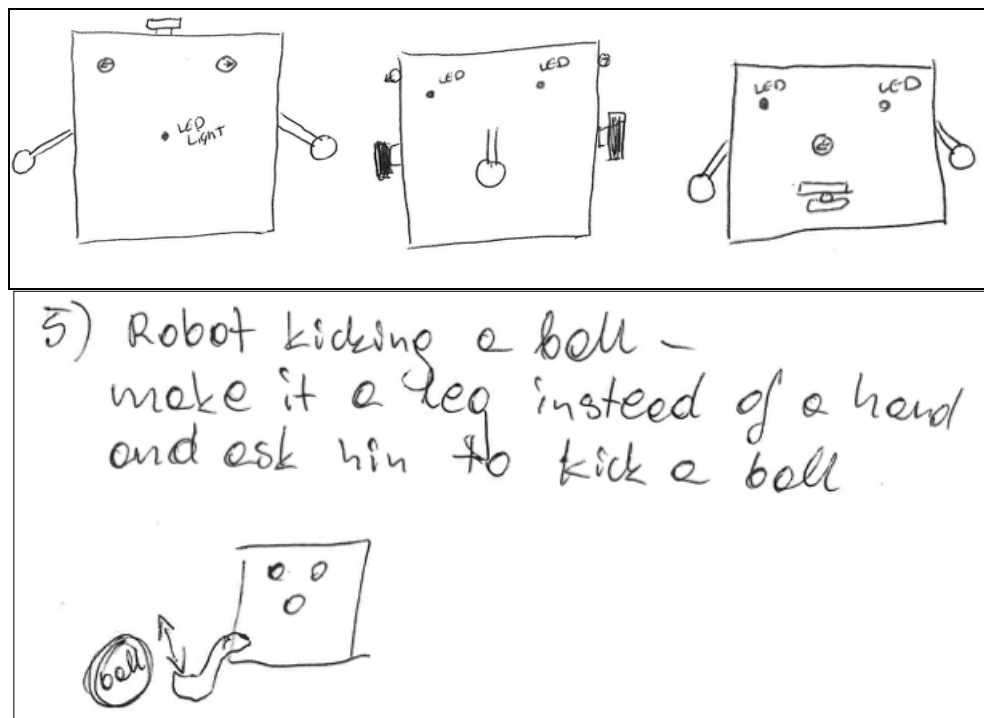


Fig. 4 One pilot participant's robot redesign designs were primarily text-based, with each focusing on one function the robot could perform.

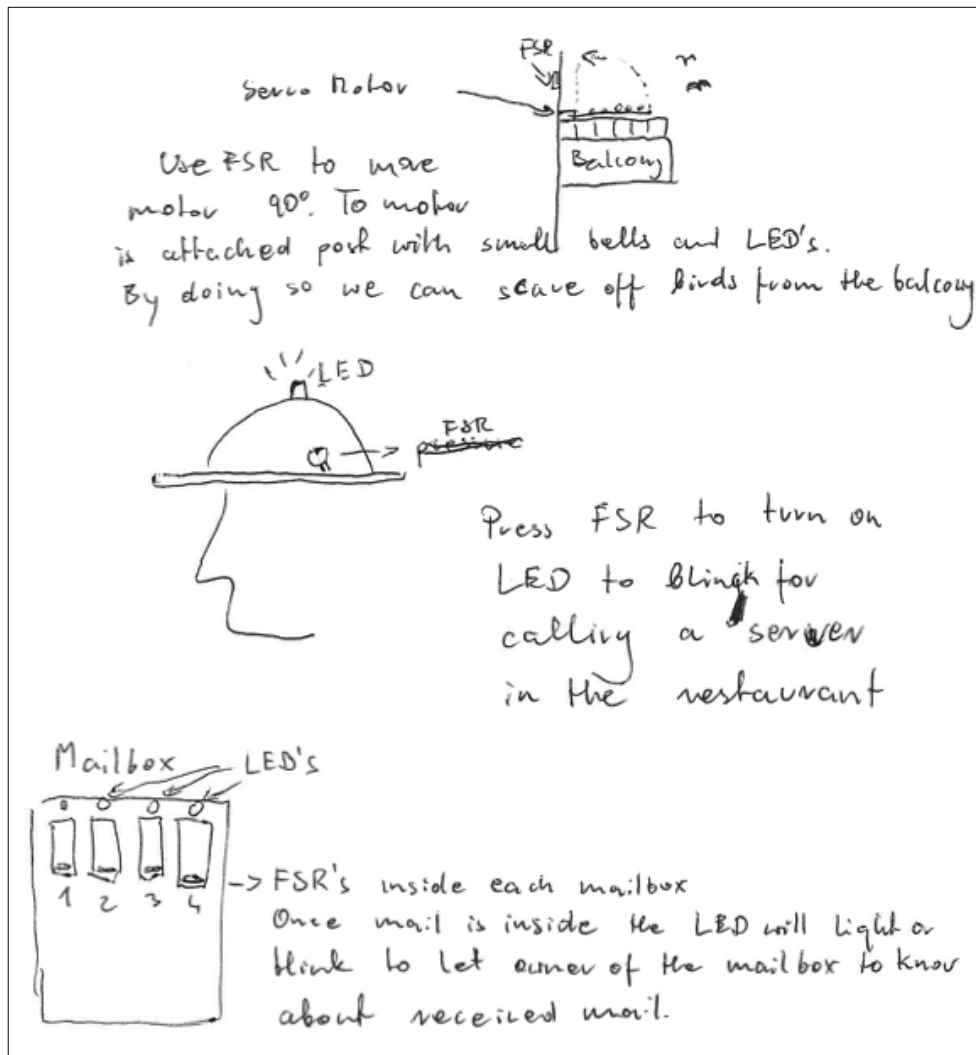


Fig. 5 One pilot participant's robot redesign sketches combined text and images, with each focusing on solving a human need in a different context.

4.4 Questionnaire Responses

Pre- and post-activity questionnaires and interviews included several notable responses. Referencing her experience building the robot after the tutorial, one participant said that she *"didn't know what was happening until I closed the lid"* of the cardboard box. Another said her favorite part of the activity was *"just seeing the robot at the very end."* Participants knew they were building a robot, given the tutorial's multiple references to *"the robot."* However, these comments suggest that as they were working through the tutorial, participants did not have a sense of the robot's final state, or that it would be noticeably different from the assembled components. This raises the question of whether to display an image or physical copy of the completed robot when participants begin the activity. In the end, we decided against this,

as doing so would have (a) focused participants on reaching an end goal rather than participating in a learning process, and (b) diminished their sense of discovery upon completing the tutorial. This also raises the question of how reflecting on steps of the tutorial or functions of the components differs from reflecting on the overall experience, and whether embodiment or asking questions can prompt the latter during different stages of system integration. To avoid interrupting the learning process, we decided against asking participants to reflect on the overall experience during the tutorial.

Regarding the difficulty of the tutorial, participants noted that it was “*easier than I imagined,*” “*wasn't as hard as I imagined,*” and “*very clear for the most part, for someone without any experience.*” They also agreed that they had learned from the two activities, describing their learnings as “*how to put together a robot, also what some components are, and how Arduinos work.*”

When asked what was most challenging about the redesign task, participants focused on ideation, including “*I wasn't sure I would have any ideas,*” “*just figuring out my ideas,*” and to “*trying to think about something to make,*” suggesting that they may yet lack confidence in their new knowledge. But even so, all participants were able to produce multiple redesign ideas.

Participants also completed Likert-style questions rating the pace, length, and difficulty of the tutorial. Pace and length ratings fell at around 4, generally corresponding to the written comments, and indicating that each was appropriate. Difficulty ratings ranged from 1–3, which is the lower end of the scale, but we consider this appropriate as the tutorial is designed to make electronics accessible to inexperienced learners, and being slightly easy is preferable to being slightly difficult. Responses also may not perfectly match experiences, as we observed participants experiencing challenges, such as remembering component names, discerning which sockets to plug components into, and responding to “*how else might you...*” questions.

Altogether, the responses and ratings suggest that the study is well matched to our participant population and timeframe without being trivial, and that it prompts learning.

5 Moving Forward

Building on the discussion of Data, Measures & Analysis above, the corpus of study data will comprise:

- Behavioral coding of participants’ physical actions and conversations, which will now focus on confirming reflective behaviors observed during pilot testing, such as lifting, holding, and testing components when asked DRQs.

- Sketches of robot designs, which we will evaluate using measures such as idea fluency (depth), flexibility (breadth), detail and completeness, focus of the robot's task (such as solving users' needs vs featuring robots' functions or form), and use of the provided components in different ways.
- Analysis of questionnaire and interview responses, including a comparison of pre- and post-activity responses to measure self-assessment of learning.

The pilot's outcomes thus far inform experiment design for future studies of agents in support of ideation and learning, as well as technology-supported education. We expect the full study's results to be relevant to the human-computer (HCI) and human-robot (HRI) interaction research communities.

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